FIBONACCI NUMBERS

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Abstract

This is an examination of the Fibonacci numbers. They are defined recursively and explicitly. A variety of propositions are proved with regard to their additivity, divisibility, etc. using methods from Calculus, Number Theory and one proof involving matrices. At the end there are a few examples of real-world applications of these numbers.

What Are Fibonacci Numbers?

Fibonacci numbers make up a sequence in which the value of the nth number in the sequence is given by the sum of the previous two numbers in the sequence. For example, the thirteenth Fibonacci number is 233 and the fourteenth Fibonacci number is 377. To obtain the fifteenth Fibonacci number, simply add $233 + 377 = 610$.

The sequence can be defined recursively as

$$F_0 = 0$$
$$F_1 = 1$$
$$F_n = F_{n-1} + F_{n-2} \quad n \geq 2$$

An explicit formula for the nth Fibonacci number can be derived from the recursive formula. It is a second order difference equation, and can be solved as follows.

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Let $F_n = x^n$. Then F_{n-1} and $F_{n-2} = x^{n-1}$ and x^{n-2} respectively. We now have

$$x^n = x^{n-1} + x^{n-2}$$

Divide all terms by x^n and multiply by x^2. Group all non-zero terms on one side to get the quadratic equation

$$x^2 - x - 1 = 0$$

The roots of this equation are

$$\frac{1 \pm \sqrt{5}}{2}$$

We can now write

$$F_n = a\left(\frac{1+\sqrt{5}}{2}\right)^n + b\left(\frac{1-\sqrt{5}}{2}\right)^n$$

To solve for a and b, choose two n values and their corresponding Fibonacci numbers and plug them in. The simplest way is to choose the values $F_0 = 0$ and $F_1 = 1$. We get

$$0 = a + b$$

$$1 = a\left(\frac{1+\sqrt{5}}{2}\right) + b\left(\frac{1-\sqrt{5}}{2}\right)$$

After solving the system of equations, we get

$$a = \frac{1}{\sqrt{5}} \qquad \text{and} \qquad b = -\frac{1}{\sqrt{5}}$$

The explicit formula for the nth Fibonacci number is known as Binet's formula and can be written

$$F_n = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right]$$

This formula can also be derived via series representation. The generating function for the Fibonacci numbers is

$$f(x) = \sum_{n=0}^{\infty} F_n x^n$$

$$= 0 + x + \sum_{n=2}^{\infty} F_n x^n$$

$$= x + \sum_{n=2}^{\infty} (F_{n-1} + F_{n-2}) x^n$$

$$= x + \sum_{n=2}^{\infty} F_{n-1} x^n + \sum_{n=2}^{\infty} F_{n-2} x^n$$

$$= x + \sum_{n=1}^{\infty} F_n x^{n+1} + \sum_{n=0}^{\infty} F_n x^{n+2}$$

$$= x + x \sum_{n=1}^{\infty} F_n x^n + x^2 \sum_{n=0}^{\infty} F_n x^n$$

Note that

$$f(x) = \sum_{n=0}^{\infty} F_n x^n = 0 + \sum_{n=1}^{\infty} F_n x^n$$

We have

$$f(x) = x + x f(x) + x^2 f(x)$$

$$1 = \frac{x}{f(x)} + x + x^2$$

$$\frac{1 - x - x^2}{x} = \frac{1}{f(x)}$$

$$\frac{x}{1 - x - x^2} = f(x)$$

The singularities of this fraction are

$$-\frac{1 \pm \sqrt{5}}{2}$$

We can rewrite the denominator in a way that allows us to write it as a power series later on. Note that

$$\frac{2}{1 + \sqrt{5}} = -\frac{1 - \sqrt{5}}{2}$$

We have

$$-\left(x + \frac{1 + \sqrt{5}}{2}\right)\left(x + \frac{1 - \sqrt{5}}{2}\right)$$

$$= -\frac{1 + \sqrt{5}}{2}\left(x + \frac{1 - \sqrt{5}}{2}\right)\left(1 - x\frac{1 - \sqrt{5}}{2}\right)$$

$$= \left(1 - x\frac{1+\sqrt{5}}{2}\right)\left(1 - x\frac{1-\sqrt{5}}{2}\right)$$

The fraction is now

$$\frac{x}{\left(1 - x\frac{1+\sqrt{5}}{2}\right)\left(1 - x\frac{1-\sqrt{5}}{2}\right)}$$

We can split it up using partial fractions

$$= \frac{\frac{1}{\sqrt{5}}}{1 - x\frac{1+\sqrt{5}}{2}} - \frac{\frac{1}{\sqrt{5}}}{1 - x\frac{1-\sqrt{5}}{2}}$$

We can write these fractions as power series

$$= \sum_{n=0}^{\infty} \frac{\left(\frac{1+\sqrt{5}}{2}\right)^n}{\sqrt{5}} X^n - \sum_{n=0}^{\infty} \frac{\left(\frac{1-\sqrt{5}}{2}\right)^n}{\sqrt{5}} X^n$$

$$= \sum_{n=0}^{\infty} \frac{\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n}{\sqrt{5}} X^n$$

If we compare this to the previous power series of

$$\sum_{n=0}^{\infty} F_n X^n$$

we have

$$F_n = \frac{\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n}{\sqrt{5}}$$

Properties of Fibonacci Numbers

It can be shown that the sum of the first n Fibonacci numbers is equal to the n+2 term minus 1. That is

$$\sum_{i=0}^{n} F_i = F_{n+2} - 1$$

To prove this, we can write the recursive Fibonacci formula as

$$F_{n+2} = F_n + F_{n+1}$$

We can rearrange the formula to get it in terms of F_n

$$F_n = F_{n+2} - F_{n+1}$$

We can begin writing the numbers in terms of this formula, and a pattern will emerge.

$$\begin{aligned} F_n &= F_{n+2} - F_{n+1} \\ F_{n-1} &= F_{n+1} - F_n \\ F_{n-2} &= F_n - F_{n-1} \\ &\vdots \\ F_1 &= F_3 - F_2 \end{aligned}$$

Adding up all the F_is will lead to cancellations on the right side. By the end we will only be left with

$$\sum_{i=1}^{n} F_i = F_{n+2} - F_2$$

$F_2 = 1$, and we have

$$\sum_{i=1}^{n} F_i = F_{n+2} - 1$$

It can be shown through induction that the sum of the first n even Fibonacci numbers is equal to

$$F_{2n+1} - 1$$

That is

$$\sum_{i=1}^{n} F_{2n} = F_{2n+1} - 1$$

The base case is simple. When $n = 1$, $F_{2n} = F_2 = F_3 - 1 = 2 - 1 = 1$.
Our inductive hypothesis is

$$F_2 + F_4 + ... + F_{2n} + F_{2n+2} = F_{2n+3} - 1$$

Assume

$$F_2 + F_4 + ... + F_{2n} = F_{2n+1} - 1$$

Add F_{2n+2} to both sides. We get

$$F_2 + F_4 + ... + F_{2n} + F_{2n+2} = F_{2n+1} + F_{2n+2} - 1$$

$F_{2n+1} + F_{2n+2} = F_{2n+3}$ by use of the recursive formula. We now have

$$F_2 + F_4 + ... + F_{2n} + F_{2n+2} = F_{2n+3} - 1$$

which was the inductive hypothesis.

Through a similar process, the sum of the first n odd Fibonacci numbers can be shown to be equal to F_{2n}. That is

$$\sum_{i=1}^{n} F_{2i-1} = F_{2n}$$

To start with, let $n = 1$. Then

$$F_{2(1)-1} = F_1 = 1 = F_2$$

The inductive hypothesis is that

$$F_1 + F_3 + ... + F_{2k-1} + F_{2k+1} = F2k + 2$$

Assume

$$F_1 + F_3 + ... + F_{2k-1} = F_{2n}$$

Add F_{2k+1} to both sides. We get

$$F_1 + F_3 + ... + F_{2k-1} + F_{2k+1} = F_{2n} + F_{2k+1}$$

Using the recursive formula for the nth Fibonacci number the right side simplifies to F_{2k+2} and we get

$$F_1 + F_3 + ... + F_{2k-1} + F_{2k+1} = F2k + 2$$

which was the inductive hypothesis.

Continuously writing out the full explicit Fibonacci formula can be cumbersome. For convenience purposes, we can let $a = \frac{1+\sqrt{5}}{2}$ and $b = \frac{1-\sqrt{5}}{2}$. Then the formula becomes

$$f_n = \frac{a^n - b^n}{\sqrt{5}}$$

Note that $ab = -1$ and $a - b = \sqrt{5}$.

Proposition: $f_{m+n} = f_m f_{n+1} + f_{m-1} f_n$

$$f_m f_{n+1} + f_{m-1} f_n = \left(\frac{a^m - b^m}{\sqrt{5}}\right)\left(\frac{a^{n+1} - b^{n+1}}{\sqrt{5}}\right) + \left(\frac{a^{m-1} - b^{m-1}}{\sqrt{5}}\right)\left(\frac{a^n - b^n}{\sqrt{5}}\right)$$

$$= \frac{1}{5}(a^{m+n+1} + b^{m+n+1} + a^{m+n-1} + b^{m+n-1}$$

$$-a^m b^{n+1} - a^{m-1} b^n - b^m a^{n+1} - b^{m-1} a^n)$$

$$= \frac{1}{5}(a^{m+n+1} + b^{m+n+1} + a^{m+n-1} + b^{m+n-1}$$

$$-a^{m+1-1} b^{n+1} - a^{m-1} b^n - b^{m+1-1} a^{n+1} - b^{m-1} a^n)$$

$$= \frac{1}{5}(a^{m+n+1} + b^{m+n+1} + a^{m+n-1} b^{1-1} + b^{m+n-1} a^{1-1})$$

$$= \frac{1}{5}(a^{m+n+1} + b^{m+n+1} - a^{m+n} b - b^{m+n} a)$$

$$= \frac{1}{5}\left[(a^{m+n} - b^{m+n})(a - b)\right]$$

$$= \frac{1}{\sqrt{5}^2}[(a^{m+n} - b^{m+n})(\sqrt{5})]$$

$$= \frac{a^{m+n} - b^{m+n}}{\sqrt{5}}$$

$$= f_{n+m}$$

This can also be proved using matrices.

Let $A=\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$. We want to show that $A^n=\begin{bmatrix} \mathrm{f}_{n-1} & \mathrm{f}_n \\ \mathrm{f}_n & \mathrm{f}_{n+1} \end{bmatrix}$

$A^1=\begin{bmatrix} \mathrm{f}_0 & \mathrm{f}_1 \\ \mathrm{f}_1 & \mathrm{f}_2 \end{bmatrix}=\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}=A$. We now assume $A^k=\begin{bmatrix} \mathrm{f}_{k-1} & \mathrm{f}_k \\ \mathrm{f}_k & \mathrm{f}_{k+1} \end{bmatrix}$

If we can show that $A^{k+1}=\begin{bmatrix} \mathrm{f}_k & \mathrm{f}_{k+1} \\ \mathrm{f}_{k+1} & \mathrm{f}_{k+2} \end{bmatrix}$, then we can prove that $A^n=\begin{bmatrix} \mathrm{f}_{n-1} & \mathrm{f}_n \\ \mathrm{f}_n & \mathrm{f}_{n+1} \end{bmatrix}$.

$$A^{k+1} = A^k A^1=\begin{bmatrix} \mathrm{f}_{k-1} & \mathrm{f}_k \\ \mathrm{f}_k & \mathrm{f}_{k+1} \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \mathrm{f}_k & \mathrm{f}_{k-1}+f_k \\ \mathrm{f}_{k+1} & \mathrm{f}_k+f_{k+1} \end{bmatrix}$$

Using the recursive formula for the nth Fibonacci number the matrix simplifies to

$$\begin{bmatrix} \mathrm{f}_k & \mathrm{f}_{k+1} \\ \mathrm{f}_{k+1} & \mathrm{f}_{k+2} \end{bmatrix}.$$

We can now use this matrix identity to prove the Fibonacci property $f_{n+m} = f_m f_{n+1} + f_{m-1} f_n$.

$A^{n+m}=\begin{bmatrix} \mathrm{f}_{n+m-1} & \mathrm{f}_{n+m} \\ \mathrm{f}_{n+m} & \mathrm{f}_{n+m+1} \end{bmatrix} = A^n A^m=\begin{bmatrix} \mathrm{f}_{n-1} & \mathrm{f}_n \\ \mathrm{f}_n & \mathrm{f}_{n+1} \end{bmatrix} \begin{bmatrix} \mathrm{f}_{m-1} & \mathrm{f}_m \\ \mathrm{f}_m & \mathrm{f}_{m+1} \end{bmatrix}$

The upper right term of the first matrix can be found by multiplying $\begin{bmatrix} \mathrm{f}_{n-1} & \mathrm{f}_n \end{bmatrix} \begin{bmatrix} \mathrm{f}_m \\ \mathrm{f}_{m+1} \end{bmatrix} = f_{n-1} f_m + f_n f_{m+1}$

$$= (f_{n+1}-f_n)f_m+f_n(f_{m-1}+f_m) = f_{n+1}f_m-f_nf_m+f_nf_m+f_nf_{m-1}$$

$$= f_m f_{n+1} + f_n f_{m-1}$$

The lower left term of the first matrix can be found by multiplying $\begin{bmatrix} \mathrm{f}_n & \mathrm{f}_{n+1} \end{bmatrix} \begin{bmatrix} \mathrm{f}_{m-1} \\ \mathrm{f}_m \end{bmatrix} = f_n f_{m-1} + f_{n+1} f_m$.

Proposition: If $m|n$ then $f_m|f_n$.

Proof: Let $m|n$. Then $n = km$, where k is some integer≥ 1. Our goal is to show that $f_m|f_{km}$.

Let $k = 1$. Then

$$f_m|f_m$$

Now assume $f_m|f_{km}$. If we can show that $f_m|f_{m(k+1)}$ then we can prove the proposition. We can use the formula we proved in the last proposition by setting $n = mk$. We have

$$f_{(mk)+m} = f_m f_{mk+1} + f_{m-1} f_{mk}$$

$f_{mk} = f_m I$ for some integer I because $f_m|f_{mk}$. We have

$$f_{(mk)+m} = f_m f_{mk+1} + f_{m-1} f_m I = f_m(f_{mk+1} + I f_{m-1})$$

f_m divides the right side of the equation, which proves the proposition.

Proposition: Two consecutive Fibonacci numbers are relatively prime. That is, the $\gcd(f_n, f_{n+1}) = 1$.

Proof: Let $k = 1$.

$$gcd(f_k, f_{k+1}) = gcd(f_1, f_2) = gcd(1, 1) = 1$$

Now let $d = gcd(f_{k+1}, f_{k+2})$.

$$d|f_{k+1} \text{ and } d|f_{k+2}$$

$$\Rightarrow d|f_{k+2} - f_{k+1}$$

$$\Rightarrow d|f_k$$

$$\Rightarrow d|gcd(f_k, f_{k+1})$$

$$\Rightarrow d|1$$

$$\Rightarrow d = 1$$

As f_n grows larger, the value of $\frac{f_{n+1}}{f_n}$ begins to approach the value of $a = \frac{1+\sqrt{5}}{2} \approx 1.618$.

Proposition:

$$\lim_{n\to\infty} \frac{f_{n+1}}{f_n} = a$$

We can prove this using the Binet formula and taking the limit as $n \to \infty$

$$\lim_{n\to\infty} \frac{\frac{a^{n+1}-b^{n+1}}{a-b}}{\frac{a^n-b^n}{a-b}}$$

After cancellation of the denominators we can split up the fraction and then split up the limits.

$$= \lim_{n\to\infty} \frac{a^{n+1}}{a^n - b^n} - \lim_{n\to\infty} \frac{b^{n+1}}{a^n - b^n}$$

Because $|b| \approx .618 > 1$, the limit on the right will become 0 and we can disregard the b^n in the denominator of the first limit because it will become infinitely smaller and will not affect the a^n term as it becomes infinitely larger.

$$= \lim_{n\to\infty} \frac{a^{n+1}}{a^n}$$

$$= \lim_{n\to\infty} a = a$$

A Brief Look at Lucas Numbers

Lucas numbers make up a recursive sequence that is defined in the same way as the Fibonacci numbers,

$$L_n = L_{n-1} + L_{n-2}$$

but with a different starting point. Lucas numbers begin with

$$L_0 = 2 \text{ and } L_1 = 1$$

as opposed to

$$F_0 = 0 \text{ and } F_1 = 1$$

Below is a table listing the first nine Fibonacci and Lucas numbers.

n	F_n	L_n
0	0	2
1	1	1
2	1	3
3	2	4
4	3	7
5	5	11
6	8	18
7	13	29
8	21	47

Proposition:

$$L_n = 2F_{n-1} + F_n \qquad n \geq 1$$

Proof: Take

$$L_k = 2F_{k-1} + F_k$$

and let $k = 1$.

$$L_1 = 2F_0 + F_1$$

$$1 = 2(0) + 1$$

$$1 = 1$$

IH:

$$L_{k+1} = 2F_k + F_{k+1}$$

We have that

$$L_k = 2F_{k-1} + F_k$$

Add L_{k-1} to both sides.

$$L_{k+1} = 2F_{k-1} + F_k + L_{k-1}$$

$$= 2F_{k-1} + F_k + 2F_{k-2} + F_{k-1}$$

$$= F_k + 3F_{k-1} + 2F_{k-2}$$

$$= F_k + 3F_{k-1} + 2(F_k - F_{k-1})$$

$$= 3F_k + F_{k-1}$$

$$= 3F_k + F_{k+1} - F_k$$

$$= 2F_k + F_{k+1}$$

Proposition:

$$F_{2n} = (F_{n+1} + F_{n-1})F_n \qquad n \geq 1$$

Proof: Let $n = 1$.

$$F_2 = (F_2 + F_0)F_1$$

$$1 = (1)1 = 1$$

Now let $n = k + 1$.

$$\text{IH:} \qquad F_{2k+2} = (F_{k+2} + F_k)F_{k+1}$$

Add F_{2k+1} to both sides of the original proposition.

$$F_{2k} + F_{2k+1} = (F_{k+1} + F_{k-1})F_k + F_{2k+1}$$

$$= F_{2k+2} = (F_{k+1} + F_{k-1})F_k + F_{2k+1}$$

Looking at the right side

$$(F_{k+1} + F_{k-1})F_k + F_{2k+1} = (F_{k+1} + F_{k-1})F_k + F_{2n} + F_{2n-1}$$

$$= (F_{k+1} + F_{k-1})F_k + F_{2k} + F_{2k} - F_{2k-2}$$

$$= 3F_n(F_{n+1} + F_{n-1} - F_{n-1}(F_n + F_{n-2}$$

$$= 3F_kF_{k+1} + 3F_kF_{k-1} - F_kF_{k-1} - F_{k-1}F_{k-2}$$

$$= 3F_kF_{k+1} + 2F_kF_{k+1} - F_{k-2}(F_{k+1} - F_k)$$

$$= 3F_{k+1}(F_{k-1} + F_{k-2}) + 2F_kF_{k-1} + F_kF_{k-2} - F_{k+1}F_{k-2}$$

$$= 3F_{k+1}F_{k-1} + 2F_{k+1}F_{k-2} + 2F_{k-1}(F_{k+2} - F_{k+1}) + F_kF_{k-2}$$

$$= F_{k+1}F_{k-1} + 2F_{k+1}F_{k-2} + 2F_{k+2}F_{k-1} + F_{k-2}(F_{k+2} - F_{k+1})$$

$$= F_{k+1}F_{k-1} + F_{k+1}F_{k-2} + 2F_{k+2}F_{k-1} + F_{k+2}(F_k - F_{k-1})$$

$$= F_{K+1}F_{k-1} + F_{k+1}F_{k-2} + F_{k+2}F_{k-1} + F_{k+2}F_k$$

$$= F_{k+1}(F_{k-1} + F_{k-2}) + F_{k+2}(F_{k-1} + F_k)$$

$$= F_{k+1}F_k + F_{k+2}F_{k+1}$$

$$= (F_{k+2} + F_k)F_{k+1}$$

Note

$$2F_{n-1} + F_n = 2F_{n-1} + F_{n+1} - F_{n-1}$$

$$= F_{k+1} + F_{k-1}$$

$$\Rightarrow F_{2n} = (F_{n+1} + F_{n-1})F_n$$

$$= L_nF_n$$

A couple of propositions involving the squares of Fibonacci numbers:

$\boldsymbol{\sum_{n=1}^{\infty} F_n^2 = F_n F_{n+1}}$

$$F_n F_{n+1} = F_n(F_n + F_{n-1})$$

$$= F_n^2 + F_{n-1}^2 + ... + F_1(F_1 + F_0)$$

$$= F_n^2 + F_{n-1}^2 + ... + F_1^2 + F_1 F_0$$

$$= F_n^2 + F_{n-1}^2 + ... + F_1^2 + 0$$

$$= F_n^2 + F_{n-1}^2 + ... + F_1^2 + F_0^2$$

$\boldsymbol{F_n^2 + F_{n+1}^2 = F_{2n+1}}$

$$F_{2n+1} = F_{2n+2} - F_{2n}$$

$$= F_{n+1}(F_{n+2} + F_n) - F_n(F_{n+1} + F_{n-1})$$

$$= F_{n+2}F_{n+1} - F_n F_{n-1}$$

$$= F_{n+1}(F_{n+1} + F_n) - F_n(F_{n+1} - F_n)$$

$$= F_{n+1}^2 + F_n^2$$

Cassini's Identity: $\boldsymbol{F_{n-1}F_{n+1} - F_n^2 = (-1)^n}$

Proof by induction:

Base case: $n = 1$

$$F_0F_2 - F_1^2 = 0 - 1 = -1 = (-1)^1$$

IH: $F_nF_{n+2} - F_{n+1}^2 = (-1)^{n+1}$

Looking at the left side

$$F_nF_{n+2} - F_{n+1}^2 = F_n(F_{n+1} + F_n) - (F_n + F_{n-1})^2$$

$$= F_nF_{n+1} - 2F_nF_{n-1} - F_{n-1}^2$$

$$= F_nF_{n+1} - 2F_{n-1}(F_{n+1} - F_{n-1}) - F_{n-1}^2$$

$$= F_n(F_n + F_{n-1}) - 2F_{n-1}F_{n+1} + F_{n-1}^2$$

$$= F_n^2 + F_nF_{n-1} - 2F_{n-1}F_{n+1} + F_{n-1}^2$$

$$= F_n^2 + F_{n-1}(F_{n+1} - F_{n-1}) - 2F_{n-1}F_{n+1} + F_{n-1}^2$$

$$= F_n^2 - F_{n-1}F_{n+1} = -(F_{n-1}F_{n+1} - F_n^2)$$

$$= -(-1)^n = (-1)^{n+1}$$

d'Ocagne's Identity: $\boldsymbol{F_mF_{n+1} - F_nF_{m+1} = (-1)^nF_{m-n}}$

Proof by (strong) induction: Let $k = n$

Base case: $k = 0$

$$F_m - 0 = F_m$$

Now assume it is true for $n = (0, 1, 2, ..., (k-1), k)$

IH: $F_m F_{k+2} - F_{k+1} F_{m+1} = (-1)^{k+1} F_{m-k-1}$

Looking at the left side

$$F_m F_{k+2} - F_{m+1} F_{k+1} = F_m(F_{k+1} + F_k) - F_{m+1}(F_k + F_{k-1})$$

Distribute and rearrange terms

$$= (F_m F_{k+1} - F_{m+1} F_k) + (F_m F_k - F_{m+1} F_{k-1})$$

We can use the assumptions for $n = k$ and $n = k - 1$ to rewrite as

$$= (-1)^k F_{m-k} + (-1)^{k-1} F_{m-k+1}$$

$$= (-1)^k (F_{m-k} - F_{m-k+1})$$

$$= -(-1)^k (F_{m-k+1} - F_{m-k})$$

$$= (-1)^{k+1} F_{m-k-1}$$

The last equality is obtained by using the recursive formula for the nth Fibonacci number.

Catalan's Identity: $\boldsymbol{F_n^2 - F_{n+r} F_{n-r} = (-1)^{n-r} F_r^2}$

Before we begin, note that

$$F_n = F_{r+n-r} = F_r F_{n-r+1} + F_{r-1} F_{n-r}$$

and

$$F_{n+r} = F_{2r+n-r} = F_{2r-1}F_{n-r} + F_{2r}F_{n-r+1}$$

$$= F_{n-r}(F_{r-1}^2 + F_r^2) + F_{n-r+1}(F_r(F_{r-1} + F_{r+1}))$$

$$= F_{n-r}(F_{r-1}^2 + F_r^2) + F_{n-r+1}(F_{r-1}F_r + F_r(F_{r-1} + F_r))$$

$$= F_{n-r}(F_{r-1}^2 + F_r^2) + F_{n-r+1}(2F_{r-1}F_r + F_r^2)$$

The left side of the identity can be written as

$$(F_{r-1}F_{n-r} + F_rF_{n-r+1})^2 - F_{n-r}[(F_{n-r}(F_{r-1}^2 + F_r^2)$$

$$+F_{n-r+1}(2F_rF_{r-1} + F_r^2)]$$

$$= F_{n-r+1}^2F_r^2 - F_{n-r}^2F_r^2 - F_{n-r}F_{n-r+1}F_r^2$$

$$= F_r^2(F_{n-r+1}^2 - F_{n-r+1}F_{n-r} - F_{n-r}^2)$$

$$= F_r^2(F_{n-r+1}(F_{n-r+1} - F_{n-r}) - F_{n-r}^2)$$

$$= F_r^2(F_{n-r+1}F_{n-r-1} - F_{n-r}^2)$$

$$= F_r^2(-1)^{n-r}$$

The last equality is obtained by applying Cassini's Identity to the term above it.

Fibonacci Numbers as a Sum of Binomial Coefficients

Consider how many ways a number n can be represented as a sum of ones and twos:
$1 = 1$
$2 = 1 + 1, 2$
$3 = 1 + 1 + 1, 2 + 1, 1 + 2$
The sum can end in either a one or a two. The number of ways n can be written as a sum ending in a one is equivalent to the number of ways $n - 1$ can be written as a sum of ones and twos. The number of ways n can be written as a sum ending in a two is equivalent to the number of ways $n - 2$ can be written as a sum of ones and twos. The total amount of ways n can be written as a sum of ones and twos is the sum of the previous two terms, the amount of ways $n - 1$ can be written added to the amount of ways $n - 2$ can be written. Lets look at $n = 4$ as an example. It can be written as a sum in five ways:

$$1 + 1 + 1 + 1, 1 + 1 + 2, 1 + 2 + 1, 2 + 1 + 1, 2 + 2$$

There are three ways the sum ends in a one, and two ways it ends in a two. That is the number of ways 3 can be written as a sum and the number of ways 2 can be written as a sum. $n = 5$ would be the sum of the number of ways to write 4 and 3, or $5 + 3 = 8$. If a_n is equal to the number of ways a number n can be written as a sum of ones and twos, then

$$a_n = a_{n-1} + a_{n-2}$$

This satisfies the Fibonacci recursive relationship. It is in fact the f_{n+1} Fibonacci number, starting with $n = 1$.
$a_1 = 1 = f_2$, $a_2 = 2 = f_3$, $a_3 = a_1 + a_2 = 3 = f_4$,
$a_n = a_{n-1} + a_{n-2} = f_{n+1}$.

n is the number we want to express as a sum. It is also the maximum number of ones that can be added together to get that number. If a two is added, it takes the place of two ones. If there are k twos in a sum, then there are $n-2k$ ones in that same sum. The maximum number of twos a group can have for an even sum is $\frac{n}{2}$. The max for an odd sum is $\lfloor \frac{n}{2} \rfloor$ because there is only one left over. If we choose k twos as k goes from 0 to $\lfloor \frac{n}{2} \rfloor$ out of all possible k twos and $n-2k$ ones, or $n-k$ terms in total, we can find the number of ways n can be represented as a sum of ones and twos and write it as a sum of binomial coefficients:

$$a_n = \sum_{k=0}^{\lfloor \frac{n}{2} \rfloor} \binom{n-k}{k}$$

Because f_{n+1} gives the number of ways a number n can be represented by a sum of ones and twos, this binomial sum is also a formula for the f_{n+1}^{th} Fibonacci number. For the sum of ones and twos, we would start at $n = 1$, but for the Fibonacci numbers we can start at $n = 0$ ($f_{0+1} = 1 = f_{1+1}$).

Real-World Uses for Fibonacci Numbers

Fibonacci numbers have relevance in stock trading in the form of Fibonacci retracements and extensions. The process is to take any two points, usually extremes, on a stock chart and multiply their difference by a Fibonacci ratio value. The values most commonly used are .236, .382, and .618 as well as 0, .5, and 1 which are not exclusive Fibonacci ratios. The values of the ratios and the percentages they correspond to are more important than the individual Fibonacci numbers in use here. $.236 \approx \frac{34}{144} = \frac{F_9}{F_{12}} \approx \frac{2584}{10949} = \frac{F_{18}}{F_{21}} \approx \frac{832040}{3524578} = \frac{F_{30}}{F_{33}}$. The 23.6% bracket can be represented by the ratio $\frac{F_n}{F_{n+3}}$. The other percentages can be represented by the ratios $\frac{F_n}{F_{n+1}}$, $\frac{F_n}{F_{n+2}}$. Fibonacci extensions are similar in calculation to retracements. The most used extension percentages are 61.8%, 100%, 161.8%, 200%, and 261.8%. The 161.8% value can be found with the ratio $\frac{F_{n+1}}{F_n}$ and 261.8% with the ratio $\frac{F_{n+2}}{F_n}$. There is no explicit formula with regards to the prices or ratios chosen.

Fibonacci numbers have relevance in computer science through a process called Fibonacci search. It is similar to binary search, but is based around the Fibonacci numbers and uses only addition and subtraction. Here is a simple example:
Consider the array of numbers $A = (2, 4, 7, 11)$. $A[0] = 2$, $A[1] = 4$, $A[2] = 7$, $A[3] = 11$. We want to find the element 7. There are 4 elements, so $n = 4$. We take the smallest Fibonacci number greater than n, which is 5. $F_m = 5$, $F_{m-1} = 3$, $F_{m-2} = 2$. Starting with the offset of -1, we take the min(offset+F_{m-2}, $n - 1$) and return the index for that number. $\text{Min}(1, 3) = 1$. $A[1] = 4 \neq 7$. We

shift the Fibonacci values down one spot and the offset becomes 1. $F_m = 3$, $F_{m-1} = 2$, $F_{m-2} = 1$. $\text{Min}(2, 3) = 2$. $A[2] = 7$. The process is now complete.

Fibonacci numbers can be used in gambling through the Fibonacci Betting System. The idea is to establish an initial betting amount, I, and multiply it by a number in the sequence, F_n. If the bet is successful, move down two places in the sequence and bet again. If the bet is unsuccessful, move up one place and bet again. The process ends when
a) The player has moved off the sequence (to F_0 or below).
b) The player has lost as much as they were willing to lose and doesn't want to risk any more.
c) The player just decides to stop.
Here is a short example.
The initial bet, I, is \$20. The starting point is $F_2 = 1$. The first bet is a loss, the next bet is $I(F_3) = \$40$. This bet is a win. The next bet is $I(F_1) = 20$. This bet and the next 3 after are losses. The next successful bet is $IF_5 = \$160$. The next two bets are successes. The player has moved off the sequence and walks away with \$60 (though they were as low as -\$160 after the 4 consecutive losses).

There are many more things that can be done with Fibonacci Numbers than I have shown here. Hopefully this has given you an appreciation for this fascinating numerical sequence.

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